

We'll divide the Arduino™ code into the following broad categories, based on the reason for its use, namely:

- ◆ Input / Output via pins
 - Digital Input / Output
 - Analog Input / Output
- ◆ Delay
- ◆ Serial Communication

Input / Output

Note: Before using a pin, it's necessary to set the function of that pin. Neglecting this can lead to unwanted behaviour from the code. By default, the pins on the ATmega based Arduino™ are configured to be used as input pins.

The function that the pin is used for is set using the 'pinMode()' function in the 'setup()' function of the code.

The pins on the Arduino™ can be used as either input pins or output pins and are configured using the following syntax:

<code>pinMode(pin, INPUT);</code>	Set pin to input pin
<code>pinMode(pin, OUTPUT);</code>	Set pin to output pin

The Arduino™ has two types of pins:

1. Digital Input/Output Pins
2. Analog Input Pins

Digital Input/Output Pins:

Digital data has only two states:

1. logic HIGH (5V)
2. logic LOW (0V).

Digital Write

To write data using the digital write pins, use the following syntax:

<code>digitalWrite(pin, HIGH);</code>	This writes a logic HIGH level to the corresponding pin. A voltage output of +5 volts is obtained if a multimeter is connected between the pin and ground.
<code>digitalWrite(pin, LOW);</code>	This writes a logic logic LOW level to the corresponding pin. A voltage output of 0 volts is obtained if a multi-meter is connected between the pin and ground.